[YOUR BRAND NAME] · ONLINE FINANCE TUTORING

FINA 3320 Exam Review

Financial Management at SMU Cox: time value of money, valuation, risk and return, capital budgeting.

PLAN

Today’s roadmap

01

Time value of money

PV, FV, annuities, perpetuities

02

Bonds and stocks

Prices, yields, dividend growth

03

Risk and return

Beta, CAPM, portfolios

04

Capital budgeting

NPV, IRR, payback

05

Cost of capital

WACC and project discount rates

06

Ratios and DuPont

Reading the financial statements

01 · TIME VALUE OF MONEY

A dollar today beats a dollar later

Future value

FV = PV × (1 + r)^n

Grow today's cash forward at rate r for n periods.

Present value

PV = FV / (1 + r)^n

Discount a future cash flow back to today.

Annuity

PV = PMT × [1 − (1 + r)^−n] / r

Level payments at the end of each period.

EXAM HABIT

Match the rate to the payment period. Monthly payments need a monthly rate and a monthly n.

Three setups to recognize on sight

Problem	Setup	Answer
$10,000 invested at 6% for 5 years	FV = 10,000 × 1.06^5	**$13,382**
$50,000 arrives in 8 years, discount at 7%	PV = 50,000 / 1.07^8	**$29,100**
$1,000 a year for 10 years at 8%	PV = 1,000 × [1 − 1.08^−10] / 0.08	**$6,710**

Answers are rounded to the nearest dollar. Check yours with the TVM keys on your calculator.

Price is the PV of coupons plus face value

Worked example

5-year bond, 6% annual coupon, $1,000 face value, 8% yield to maturity.

P = 60 × [1 − 1.08^−5] / 0.08 + 1,000 / 1.08^5

P ≈ $239.56 + $680.58 = $920.15

Coupon vs. yield

If	Bond sells at
Coupon rate < YTM	Discount
Coupon rate = YTM	Par
Coupon rate > YTM	Premium

Rates up, prices down. Longer maturity means more price sensitivity.

Growing dividends, constant growth

$P_0 = D_1 / (r - g)$

$D_1 = D_0 \times (1 + g)$. The model needs g below r, and it prices the stock one period before D_1 arrives.

Example: $D_1 = \$2.00$, $r = 10\%$, $g = 4\%$.

STOCK PRICE

$$\$33.33$$

$2.00 / (0.10 - 0.04)$

CAPM prices only market risk

Security market line

E(R) = rf + β × (E(Rm) − rf)

Example: rf = 4%, β = 1.2, market risk premium = 6%.

E(R) = 4% + 1.2 × 6% = 11.2%

Systematic risk

Market-wide, so it cannot be diversified away.

Portfolio beta

Weighted average of the stocks' betas.

Three decision rules, one that matters most

Method	Accept if	Watch out
NPV	NPV is greater than 0	Gives value added in dollars, so it is the benchmark.
IRR	IRR exceeds the required return	Can mislead on mutually exclusive projects or sign changes.
Payback	Payback is within the cutoff	Ignores the time value of money and later cash flows.

When NPV and IRR disagree on mutually exclusive projects, follow NPV.

04 · WORKED EXAMPLE

NPV at a 10% discount rate

Year	Cash flow	Present value
0	−$100,000	−$100,000
1	$40,000	$36,364
2	$45,000	$37,190
3	$50,000	$37,566

NPV

$11,120

Positive, so accept the project.

WACC blends debt and equity

Formula

WACC = (E/V) × Re + (D/V) × Rd × (1 − Tc)

Use market values for the weights. Debt is cheaper after the interest tax shield.

Example

60% equity at 12%, 40% debt at 6% pre-tax, tax rate 25%.

0.60 × 12% + 0.40 × 6% × 0.75

= 7.2% + 1.8% = 9.0%

DuPont splits ROE into three drivers

Profitability

Profit margin

Net income / Sales

×

Efficiency

Asset turnover

Sales / Total assets

×

Leverage

Equity multiplier

Total assets / Equity

ROE = Profit margin × Asset turnover × Equity multiplier. Current ratio = Current assets / Current liabilities.

EXAM STRATEGY

Six common mistakes

Rate and period

Annual rate with monthly payments.

Wrong dividend

Using D0 where D1 belongs.

Risk premium

Treating E(Rm) as the premium.

Sunk costs

Past spending does not enter NPV.

Pre-tax debt

Forgetting (1 − Tc) in WACC.

Calculator mode

Leaving BGN on or old TVM values in.

NEXT STEPS

Turn this review into practice

1

Redo every worked example without looking.

2

Work past homework and exam problems, timed.

3

Book a session on your weakest topic: [booking link].

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